

2016 量子 725

一、

光场的粒子性：光电效应、黑体辐射、康普顿散射

能量交换光谱：弗朗克—赫兹实验

粒子的波动性：戴维孙—革末实验

二、

费米子： $\psi_A = \frac{1}{\sqrt{2}} \psi_m(x_1) \psi_n(x_2) [\chi_{\uparrow}(1) \chi_{\downarrow}(2) - \chi_{\downarrow}(1) \chi_{\uparrow}(2)]$

$m \neq n$ $\psi_A = \frac{1}{2} [\psi_m(x_1) \psi_n(x_2) + \psi_n(x_1) \psi_m(x_2)] [\chi_{\uparrow}(1) \chi_{\downarrow}(2) - \chi_{\downarrow}(1) \chi_{\uparrow}(2)]$

或 $\psi_A = \frac{1}{\sqrt{2}} [\psi_m(x_1) \psi_n(x_2) - \psi_n(x_1) \psi_m(x_2)] \begin{cases} \chi_{\uparrow}(1) \chi_{\downarrow}(2) \\ \chi_{\downarrow}(1) \chi_{\uparrow}(2) \\ \frac{1}{\sqrt{2}} [\chi_{\uparrow}(1) \chi_{\downarrow}(1) + \chi_{\downarrow}(1) \chi_{\uparrow}(2)] \end{cases}$

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$m \neq n$ $\psi_A = \frac{1}{2} [\psi_m(x_1) \psi_n(x_2) + \psi_n(x_1) \psi_m(x_2)] [\chi_{\uparrow}(1) \chi_{\downarrow}(2) - \chi_{\downarrow}(1) \chi_{\uparrow}(2)]$

或 $\psi_A = \frac{1}{\sqrt{2}} [\psi_m(x_1) \psi_n(x_2) + \psi_n(x_1) \psi_m(x_2)] \begin{cases} \chi_{\uparrow}(1) \chi_{\uparrow}(2) \\ \chi_{\downarrow}(1) \chi_{\downarrow}(2) \\ \frac{1}{\sqrt{2}} [\chi_{\uparrow}(1) \chi_{\downarrow}(1) + \chi_{\downarrow}(1) \chi_{\uparrow}(2)] \end{cases}$

三、

(1) $P = \int_0^{\pi} \int_0^{2\pi} |\psi(r, \theta, \varphi, t)|^2 \sin \theta d\theta d\varphi r^2 dr$

(2) $P = \int_0^{\infty} |\psi(r, \theta, \varphi, t)|^2 r^2 dr d\Omega$

四、

(1) $\psi_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $\psi_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\psi_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$E_1 = 0$

$E_2 = \hbar\omega$

$E_3 = 2\hbar\omega$

(2) $H_{11} = \psi_1^\dagger \lambda H' \psi_1$ $H_{12} = \psi_1^\dagger \lambda H' \psi_2$ $H_{13} = \psi_1^\dagger \lambda H' \psi_3$

$H_{21} = \psi_2^\dagger \lambda H' \psi_1$ $H_{22} = \psi_2^\dagger \lambda H' \psi_2$ $H_{23} = \psi_2^\dagger \lambda H' \psi_3$

$H_{31} = \psi_3^\dagger \lambda H' \psi_1$ $H_{32} = \psi_3^\dagger \lambda H' \psi_2$ $H_{33} = \psi_3^\dagger \lambda H' \psi_3$

微扰成立条件 $\left| \frac{H_{mn}}{E_n - E_m} \right| \ll 1$

五、

(1) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$

(2) S_x 本征函数为 $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\therefore P_{\frac{1}{2}} = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2}$

$P_{-\frac{1}{2}} = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2}$

(3) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$

(4) S_n 本征态 $\chi_+ = \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{i\phi}{2}} \\ \sin \frac{\theta}{2} e^{\frac{i\phi}{2}} \end{pmatrix}$ $\chi_- = \begin{pmatrix} -\sin \frac{\theta}{2} e^{-\frac{i\phi}{2}} \\ \cos \frac{\theta}{2} e^{\frac{i\phi}{2}} \end{pmatrix}$

相应概率为 $P_+ = \sin^2 \frac{\theta}{2}$ $P_- = \cos^2 \frac{\theta}{2}$

$\therefore \langle S_n \rangle = \frac{1}{2} (\sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2}) = -\frac{1}{2} \cos \theta$

六.

(1) 力学量完全集 (H, L^2, L_z)

本征函数为 $Y_{lm}(\theta, \varphi)$ $l=0, 1, 2, \dots$ $m=0, \pm 1, \dots, \pm l$

简并度为 $2l+1$ $E = \frac{l(l+1)\hbar^2}{2I}$

(2) $E_0 = 0$ $\psi_0 = Y_{00} = \frac{1}{\sqrt{\pi}}$ 无简并

$$E_0^{(1)} = H_{00}' = \iint Y_{00}^* H' Y_{00} d\Omega = \frac{A}{4\pi} \int_0^\pi \sin\theta d\theta \int_0^{2\pi} \cos\varphi d\varphi = 0$$

$$\therefore E_1 = 0 \quad \psi^{(0)} = \psi_0 = Y_{00} = \frac{1}{\sqrt{\pi}}$$

第一激发态 $E_1^{(0)} = \frac{3\hbar^2}{2I} = \frac{3}{2} E_0$ 三重简并

令 $\varphi_1 = Y_{11}$ $\varphi_2 = Y_{10}$ $\varphi_3 = Y_{1,-1}$

设 $\psi^{(1)} = c_1 \varphi_1 + c_2 \varphi_2 + c_3 \varphi_3$ 得

$$\begin{pmatrix} H_{11}' - E^{(1)} & H_{12}' & H_{13}' \\ H_{21}' & H_{22}' - E^{(1)} & H_{23}' \\ H_{31}' & H_{32}' & H_{33}' - E^{(1)} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = 0 \quad \text{其中 } H_{ij}' = \int \varphi_i^* H' \varphi_j d\Omega$$

得 $H_{13}' = H_{31}' = -\frac{A}{2}$, 其余为0

$$\text{即 } \begin{pmatrix} -E^{(1)} & 0 & -\frac{A}{2} \\ 0 & -E^{(1)} & 0 \\ -\frac{A}{2} & 0 & -E^{(1)} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = 0 \quad \det I = 0$$

$$\Rightarrow E_1^{(1)} = -\frac{A}{2} \quad E_2^{(1)} = 0 \quad E_3^{(1)} = \frac{A}{2}$$

$$\psi_1^{(1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \psi_2^{(1)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \psi_3^{(1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$\therefore$ 一级近似能量和零级近似波函数为

$$E_1 = \frac{3\hbar^2}{2I} - \frac{A}{2} \quad \psi_1^{(0)} = \frac{1}{\sqrt{2}} (Y_{11} + Y_{1,-1})$$

$$E_2 = \frac{3\hbar^2}{2I} \quad \psi_2^{(0)} = Y_{10}$$

$$E_3 = \frac{3\hbar^2}{2I} + \frac{A}{2} \quad \psi_3^{(0)} = \frac{1}{\sqrt{2}} (Y_{11} - Y_{1,-1})$$